

Report 1: ECG Heartbeat Categorization

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Abstract

This report presents a 1D Convolutional Neural Network (1D-CNN) for classifying ECG heartbeats using two public datasets: MIT-BIH Arrhythmia (5 classes) and PTB Diagnostic (2 classes). The model achieves 97.24% accuracy on MIT-BIH and 91.34% on PTBDB after 5 training epochs.

1 Introduction

Electrocardiogram (ECG) signal classification is important for detecting heart abnormalities. This report implements a deep learning approach using 1D-CNN to automatically classify heartbeats from ECG signals.

2 Dataset Description

The study uses two datasets from Kaggle:

2.1 MIT-BIH Arrhythmia Dataset

- **Task:** Multi-class classification (5 types of heartbeats)
- **Training samples:** 87,554
- **Test samples:** 21,892
- **Class distribution (training set):**
 - N (Normal): 72,471 samples (82.8%)
 - S: 2,223 samples (2.5%)
 - V: 5,788 samples (6.6%)
 - F: 641 samples (0.7%)
 - Q: 6,431 samples (7.3%)

2.2 PTB Diagnostic Dataset

- **Task:** Binary classification (Normal vs Abnormal)
- **Total samples:** Combined from normal and abnormal files
- **Split:** 80% training, 20% testing

All ECG signals have 187 time-steps, sampled at 125Hz (approximately 1.5 seconds per sample).

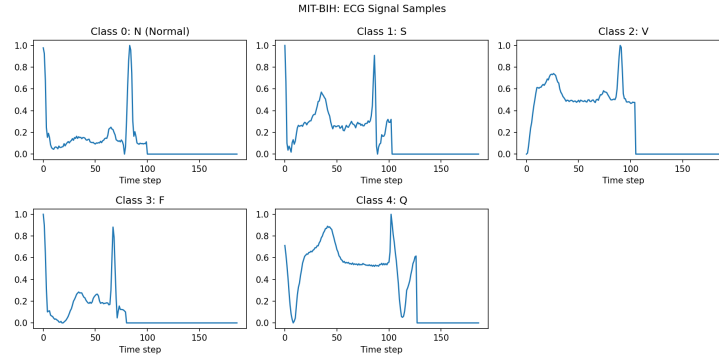


Figure 1: Sample ECG signals from each class in MIT-BIH dataset

3 Methodology

3.1 1D-CNN Architecture

The model is designed to automatically learn features from raw ECG signals:

- **Input layer:** 187 time-steps $\times$ 1 channel
- **Conv1D layer 1:** 64 filters, kernel size 6, ReLU activation
- **BatchNormalization:** Stabilize training
- **MaxPooling1D:** Pool size 3, stride 2
- **Conv1D layer 2:** 64 filters, kernel size 3, ReLU activation
- **BatchNormalization + MaxPooling1D**
- **Flatten + Dense (64 units) + Dropout (0.2)**
- **Output layer:** Softmax (5 units for MIT-BIH, 2 units for PTBDB)

3.2 Training Setup

- **Optimizer:** Adam
- **Loss function:** Categorical crossentropy
- **Epochs:** 5
- **Batch size:** 64

4 Experiments and Results

4.1 MIT-BIH Results

Training progress over 5 epochs:

- Epoch 1: 93.33% $\rightarrow$ 96.63%
- Epoch 2: 97.25% $\rightarrow$ 96.27%
- Epoch 3: 97.86% $\rightarrow$ 97.87%
- Epoch 4: 98.12% $\rightarrow$ 97.76%
- Epoch 5: 98.36% $\rightarrow$ 97.24%

Metric	Value
Training Accuracy	98.39%
Validation Accuracy	97.24%

Table 1: MIT-BIH final results

4.2 PTBDB Results

Training progress over 5 epochs:

- Epoch 1: 80.62% $\rightarrow$ 72.21%
- Epoch 2: 93.40% $\rightarrow$ 72.24%
- Epoch 3: 96.78% $\rightarrow$ 85.95%
- Epoch 4: 97.10% $\rightarrow$ 94.81%
- Epoch 5: 97.91% $\rightarrow$ 91.34%

Metric	Value
Training Accuracy	98.08%
Validation Accuracy	91.34%

Table 2: PTBDB final results

5 Comparison with Original Paper

According to Kachuee et al. (2018):

- MIT-BIH benchmark: 98% → Our model: **97.24%** (comparable)
- PTBDB benchmark: 95.9% → Our model: **91.34%** (slightly lower)

The lower result on PTBDB may be due to:

- Limited training data
- Need for hyperparameter tuning
- Different data preprocessing methods

6 Conclusion

This project successfully implemented a 1D-CNN for ECG heartbeat classification. Key findings:

- The model performs well on MIT-BIH dataset (97.24% accuracy)
- Results on PTBDB (91.34%) show room for improvement
- Data imbalance in MIT-BIH (class F only 0.7%) did not significantly affect results
- Future work: experiment with more epochs, different architectures, and data augmentation

7 Citation

Li, X., Zhang, F., Sun, Z., Li, D., Kong, X., & Zhang, Y. (2021). Automatic heartbeat classification using S-shaped reconstruction and a squeeze-and-excitation residual network. *Computers in Biology and Medicine*, 140, 105108. <https://doi.org/10.1016/j.combiomed.2021.105108>